
D4.6 - Research Agenda activity report

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1 Research Agenda activity report

1.1 Disclaimer

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Words displayed in red indicate hyperlinks. These links can be clicked or viewed by hovering over the text.

1.2 Distribution and License

The document is distributed under Creative Common Attribution 4.0 International [CC-BY](#).

The document is distributed as TLP:CLEAR.

1.3 Deliverable Definition

The identifier of the deliverable is D4.6 and it adheres to the definition outlined in the grant agreement written in bold. **Research Agenda activity report**. The description is: **Projects thesis and internship reports**.

1.4 Executive Summary

This report documents the internships and student research activities hosted by NGSOTI partners over the course of the project, as a concrete output of the project's Research Agenda. The Restena Foundation, CIRCL, and the University of Luxembourg hosted or supported students and trainees on focused technical projects that build directly on NGSOTI tools and data sources. The activities range from full Master's theses combining data analysis and machine learning, to short, hands-on engineering internships producing working prototypes and code contributions to existing open source tools.

Each activity is reported using the following structure for consistency: host institution, academic affiliation, period, context and methodology, and key outcomes. Altogether, the activities reported in the present report cover four of the project's core technical areas: passive network traffic analysis for threat intelligence, security scanning of shared infrastructure, and vulnerability intelligence tooling and machine learning support for CTI platforms.

1.5 Internships and theses hosted by the Restena Foundation

1.5.1 Lucas Villière: Final destination blackhole! Building Threat intel from trash!

Host institution: Restena Foundation

Academic affiliation: Master in Information and Computer Sciences (MICS), University of Luxembourg

Period: Master's thesis internship, submitted August 2025

Context and Methodology

Restena operates a network telescope, a passive sensor capturing all traffic sent to an inactive IP space. Existing supervised classifiers, trained on simulated ICS attacks or the labelled CIC-Darknet2020 dataset, do not transfer to this traffic: blackhole captures carry no reliable ground truth, are mostly noise, and drift over time. The thesis instead framed the problem as semi-supervised classification. Only UDP packets were kept, since TCP traffic to an inactive destination never completes a handshake, and flows were aggregated using a zero-timeout strategy to produce the flat feature vectors the classifier needs. Each flow was vectorised into close to 190 dimensions, combining network metadata with TF-IDF and Singular Value Decomposition (SVD) features extracted from the payload. Daily capture volumes, several million flows once vectorised, were processed on the University of Luxembourg's Iris HPC cluster. A Random Forest classifier was then trained without ground truth on flows aggregated over 12-hour windows, found to outperform 24-hour aggregation for surfacing anomalous patterns. As a concrete illustration of the kind of signal the approach surfaces, the intern traced a recurring UDP stream on port 514 (syslog) to an internal monitoring system at a European university, leaking ATK_ATTACK_REPORT alerts naming an internal source IP and an external Microsoft Corporation IP, evidence of a misconfigured logging destination exposed to the open Internet.

Key Outcomes

- **Structured Dataset:** Produced a novel dataset built for semi-supervised classification of UDP blackhole traffic.
- **Scalable Pipeline:** Established an HPC-compatible pipeline turning millions of raw daily packets into vectorised, ~190-dimension threat indicators.
- **Optimised Detection Window:** Trained a Random Forest model showing that 12-hour flow aggregation outperforms 24-hour aggregation for surfacing anomalous patterns.
- **Actionable Intelligence:** Isolated a concrete misconfiguration, tracing a recurring UDP stream on port 514 back to a European university leaking internal security alerts onto the open Internet.

1.5.2 Sam Kafaï: URL Checker Tools for edu.lu

Host institution: Restena Foundation

Academic affiliation: 1st year, BTS CyberSecurity, LGK

Period: Two-week internship, autumn 2025

Context and Methodology

Restena hosts edu.lu, a URL shortener for the Luxembourgish educational community built on the open source SURFshort backend. Because shortened URLs are a common vector for phishing, malware, and command-and-control, the internship built a security scanning layer that screens a target before edu.lu trusts it. The result is a modular Python CLI organised around an inheritance-based `BaseProvider` class: adding a new threat-intelligence source means writing one class, with no changes to the CLI itself. A scan runs a target through three stages: provider reputation queries, content scanning with YARA rules and redirect-chain inspection, and a unified scoring and confidence calculation. Nine providers were integrated, each with a documented role: VirusTotal as the primary broad-coverage check, Google Safe Browsing as narrower but reliable, WHOIS as a fast baseline, Whalebone for DNS-based categorisation, MISP for cross-referencing and reporting, local YARA rules as a fallback when external providers are unreachable, and Lookyloo and URLScan.io, evaluated but downweighted for being slow and inconsistent. An automation mode (`--robot`) writes a human-readable log and a detailed provider-by-provider trace per session, with MISP event IDs embedded for traceability. Run against the EICAR test file, the tool scored it 79/100 and MALICIOUS, flagged by MISP, Whalebone, VirusTotal, and Google Safe Browsing, and automatically opened a corresponding MISP event.

Key Outcomes

- **Functional Security Scanner:** Deployed a working CLI tool integrating nine threat-intelligence providers with unified scoring and confidence metrics.
- **Automated Threat Reporting:** Demonstrated end-to-end detection on a live malicious-URL test case, with automatic MISP event creation.
- **Provider Evaluation:** Produced a documented, evidence-based comparison of provider strengths and weaknesses to guide future investment in edu.lu's scanning layer.

1.6 Internships and theses hosted by CIRCL

1.6.1 Thomas Lacroix: Reinforcement and modernization of the MISP open source application

Host institution: CIRCL

Academic affiliation: TELECOM Nancy (INP Lorraine, Université de Lorraine), final-year engineering internship

Period: Academic year 2024-2025

Context and Methodology

MISP, the threat intelligence sharing platform underpinning much of NGSOTI's CTI tooling, needed both feature work on its core application and a way to verify that instance-to-instance synchronization keeps working as the codebase evolves. On the application side, the intern added filtering for extended events on both the index view and the REST API, added visual distinction and filtering for the two warning-list categories (false positive versus known identifier), and introduced a per-role cap on REST API result counts to close a denial-of-service path. Smaller fixes followed: Accept-header fallback for response format, a -k flag on the auto-generated curl command when SSL validation is skipped, a shortcut to import attributes directly from a file, and an attribute count added to the galaxy cluster view. On the synchronization side, CIRCL had no test environment for this: the intern built a Forgejo Actions pipeline running inside LXC containers, ported from the project's existing GitHub Actions CI, and a deployment script that spins up several MISP instances locally and wires them together according to a specified topology. Using this setup, a seven-instance test topology was built covering event, sighting, analyst data, and galaxy cluster propagation via PUSH and PULL, modification and deletion of synced data, tag and galaxy scoping, distribution-level changes, sharing groups, and the rules specific to internal (same-organisation) instances.

Key Outcomes

- **Core Code Contributions:** Merged usability, security, and traceability improvements into `misp-core`, including extended-event filtering, warning-list filtering, and per-role API limits.
- **Automated CI Infrastructure:** Delivered the project's first Forgejo Actions-based synchronization test suite, with reusable multi-instance deployment tooling.
- **Test Coverage:** Built a seven-instance topology validating the main synchronization scenarios across PUSH, PULL, and sharing-group rules.

Useful links:

- <https://www.misp-project.org/2025/09/19/misp-synchronisation.html/>
- Thesis: https://www.misp-project.org/Internship-report_Lacroix.pdf

1.6.2 Léa Ulasan: Vulnerability intelligence tooling (vulnerability-lookup and VulnTrain)

Host institution: CIRCL c/o Luxembourg House of Cybersecurity

Academic affiliation: ECE Engineering School

Period: May 2025 to September 2025

Context and Methodology

CIRCL maintains **Vulnerability-Lookup**, a platform for searching and correlating CVE, CWE, and related advisories, and **VulnTrain**, a companion project that trains machine learning models on vulnerability

data. The internship surfaced CWE intelligence directly in vulnerability-lookup's interface and built a pipeline predicting CWE categories from free-text vulnerability descriptions. The internship focused on extending the Vulnerability-Lookup platform with improved support for CWE-based vulnerability analysis. Vulnerability-Lookup already provides a structured environment to collect, correlate and consult vulnerability information from multiple sources. The work carried out during the internship aimed to enrich this environment by introducing dedicated Common Weakness Enumeration (CWE) views, statistics and navigation features. The methodology followed an incremental development approach: initial templates and backend blueprints were created, followed by user interface improvements, API endpoint preparation, statistical visualisations, filtering capabilities, and enrichment of CWE pages with related CVEs, mitigations and CAPEC attack-pattern information. The implementation was progressively refined through fixes, layout changes and typing improvements to ensure better usability and maintainability.

Key Outcomes

AI models:

- **Severity classification of vulnerabilities with description written in Chinese:** A RoBERTa-Based Model for Automated Vulnerability Severity Classification for Chinese. The model is available on [Hugging Face](#).
- **CWE guessing:** Predict CWE categories from Git commit messages and vulnerability descriptions. The model is available on [Hugging Face](#) as well as the [dataset with the patches](#).

New features in Vulnerability-Lookup:

- **CWE functionality:** Added dedicated CWE statistics functionality, including visualisation of the most frequent CWE categories and time-based filtering capabilities.
- **CVE and CWE visualisation:** Improved the connection between CVEs and CWEs by displaying associated CWEs on vulnerability pages and linking CWE entries to related CVEs.
- **Enhanced the usability:** of Vulnerability-Lookup through layout improvements, filtering options, API endpoint preparation and fixes to ensure clearer analyst-facing views.

1.6.3 Adrian Maraj: Lookyloo – Phishing Detection

Host institution: CIRCL co Luxembourg House of Cybersecurity

Academic affiliation: ISFATES

Period: Internship, March 2024 to August 2024

Context and Methodology

CIRCL maintains [Lookyloo](#), is a web interface for capturing a website page and mapping its browsing journey as a tree of domains, redirects, and external resources, which makes it useful for phishing and suspicious-website analysis. In the context of phishing detection, the work aimed to improve how

suspicious URLs can be submitted, captured, categorised and reviewed by analysts. The methodology followed an incremental software development approach, starting from practical improvements to the capture and reporting workflow and progressively adding features that support operational use. This included work on simplified URL submission, automatic reporting, private captures, API-based access to categorised captures, and usability improvements such as downloadable capture trees. These contributions helped make the phishing-analysis workflow more accessible, structured and suitable for repeated operational use.

Key Outcomes

- **Workflow improvement:** Improved the suspicious URL submission workflow by contributing to a simplified capture interface for phishing analysis.
- **Automatic reporting:** Added support for automatic reporting of captured URLs, helping streamline the handling of suspicious websites.
- **Operational control:** Enhanced operational control by supporting private captures and requiring authentication for selected capture workflows.
- **Analyst usability:** Improved analyst usability by adding features such as downloadable capture trees and API access to captures by category.

1.6.4 Antonia Koch: Lookyloo – Phishing Detection

Host institution: CIRCL co Luxembourg House of Cybersecurity

Academic affiliation: ISFATES

Period: March 2024 to August 2024

Context and Methodology

The internship focused on contributing to Lookyloo. Antonia's work supported the improvement of analyst-facing reporting and classification features, making it easier to process suspicious captures and prepare them for operational review. The methodology followed an incremental software development approach, combining frontend improvements, reporting workflow enhancements and API-related functionality. Contributions included improvements to the capture page, report form, administrative menus, auto-reporting options, capture upload via API, quick categorisation features, and WHOIS lookup handling. These developments helped strengthen the usability and operational efficiency of Lookyloo for phishing investigation and suspicious website analysis.

Key Outcomes

- **Reporting workflow:** Improved the phishing reporting workflow by enhancing the capture page and making the report form easier to use.
- **Operational features:** Added operational features for analysts and administrators, including auto-reporting options and improved admin-only controls.

- **Efficiency on Capture related data:** Contributed to API-based functionality, including support for uploading captures and handling capture-related data more efficiently.
- **Streamlining enrichment:** Improved classification and enrichment capabilities through quick categorisation features and better handling of WHOIS lookup information.

1.6.5 Théo Geffe: Rulezet

Host institution: CIRCL co Luxembourg House of Cybersecurity

Academic affiliation: UFR MIM Metz, Université de Lorraine

Period: April 2025 to July 2025

Context and Methodology

The internship focused on the development of **Rulezet Core**, an open-source platform for managing, validating and sharing cybersecurity detection rules across formats such as YARA, Sigma, Suricata, Zeek, Wazuh and Elastic. In the context of detection engineering, the work aimed to strengthen the platform as a collaborative environment where analysts can import rules, organise them, validate their syntax, search them, and use them in operational or training contexts. The methodology followed an incremental software engineering approach, with contributions implemented directly in the main codebase through iterative commits. The work covered backend and frontend development, rule management features, validation workflows, API-related functionality, documentation, configuration, and general maintainability improvements. This helped consolidate Rulezet as a practical tool for detection-rule lifecycle management and cybersecurity training.

Key Outcomes

- **Detection-rule management:** Extended Rulezet's detection-rule management capabilities, supporting the handling of multiple rule formats used by SOC analysts and detection engineers.
- **Rule validation:** Improved rule validation and import workflows, helping users assess the quality and usability of detection rules before operational use.
- **Usability:** Contributed to platform usability and maintainability through frontend refinements, backend improvements, configuration updates and documentation work.
- **Collaborative Detection Engineering:** Strengthened Rulezet as an open-source cybersecurity platform suitable for collaborative detection engineering, rule sharing, and training activities.

1.7 Thesis supported by the University of Luxembourg

1.7.1 Njomza Rexhepi: Understanding Network Telescope Traffic Through Exploratory Data Analysis and Transformer-Based Representation Learning

Host institution: University of Luxembourg, Interdisciplinary Centre for Security, Reliability and Trust (SnT)

Academic affiliation: Master's programme, Faculty of Electrical and Computer Engineering, University of Prishtina, with support from LuxDev (Luxembourg Development Agency)

Period: Master's thesis in progress, expected submission September 2026 [Ongoing]

Context and Methodology This thesis investigates how transformer-based representation learning, the approach behind models like ET-BERT for encrypted traffic, can be adapted to a setting it was not designed for: passive network telescope data, where traffic is unsolicited, single-direction, and largely without application content. The intern built a full-year extraction pipeline from raw telescope captures and used it to characterize two traffic types reaching the telescope, Telnet scanning and DNS queries, as a basis for understanding what these models would actually be learning from. That characterization produced a more nuanced picture than initially assumed: known botnet signatures (Mirai-family) are present but represent a minority of scanning traffic, and DNS traffic at the telescope is dominated by misdirected legitimate queries rather than attacks, with malicious activity forming a smaller, distinct subset. An exploratory look at whether the same sources show correlated TCP and DNS activity over time surfaced candidate patterns worth following up, though not yet conclusive evidence of coordinated behavior. Because the dominant traffic carries no payload to tokenize in the way ET-BERT originally assumes, the intern is now adapting that architecture's tokenization and pretraining approach to work directly from packet header structure instead.

Key Outcomes

- **Data Extraction Pipeline:** Delivered a cluster-scale pipeline converting a full year of raw telescope PCAPs into a structured, schema-consistent Parquet dataset, with totals cross-validated across multiple independent analysis scripts.
- **Behavioral Characterization:** Produced a full-year profiling of Telnet scanning activity and DNS query composition, distinguishing the minority of traffic attributable to known botnet/scanner tooling from the majority that is either unattributed or, in the DNS channel, benign misdirected traffic.
- **Cross-Protocol Exploration:** Identified candidate IP cohorts and subnets showing temporal co-activity across the TCP and DNS channels, currently framed as exploratory findings pending statistical and threat-intelligence validation.
- **Architectural Adaptation:** Established structure-based tokenizers and an ET-BERT adaptation for single-sided, payload-sparse traffic, with pretraining and evaluation ongoing ahead of the September 2026 submission.

1.8 Appendix

1.8.1 Internships documents and outcomes

- Lucas Villière, *Final destination blackhole! Building Threat intel from trash!*, Master's thesis, University of Luxembourg, August 2025.

- Sam Kafai, Cédric Renzi, Denim Latic, *URL Checker Tools for edu.lu (NGSOTI project)*, presentation, OpenSource Conference 2025.
- Thomas Lacroix, *Reinforcement and modernization of the MISP open source application*, Master's thesis, TELECOM Nancy, 2025, https://www.misp-project.org/Internship-report_Lacroix.pdf.
- Njomza Rexhepi, *Understanding Network Telescope Traffic Through Exploratory Data Analysis and Transformer-Based Representation Learning*, Master's thesis (in progress), University of Prishtina, with support from the University of Luxembourg (SnT) and LuxDev, expected September 2026.
- Léa Ulsan. *Vulnerability Management and Threat Intelligence*, ECE Engineering School, Git commits during internship, 2025, <https://github.com/vulnerability-lookup/vulnerability-lookup/commits?author=hamartia.eu%40gmail.com>
- Adrian Maraj, *Lookyloo Phishing Detection*. ISFATES, Git commits during internship, 2024, <https://github.com/Lookyloo/lookyloo/commits?author=adrima01>
- Antonia Koch, *Lookyloo Phishing Detection*. ISFATES, Git commits during internship, 2024, <https://github.com/Lookyloo/lookyloo/commits?author=AntoniaBK>
- Théo Geffe, **Rulezet**, UFR MIM Metz, Université de Lorraine, Git commits during internship, 2025, <https://github.com/rulezet/rulezet-core/commits/main/?since=2025-04-07&until=2025-07-25&author=ecrou-exact>
- Presentation at BSides Luxembourg from Léa Ulsan, 2025-06-19, "**When Data Talks, We Let AI Listen**"